Lecture 4. Formal Introduction to Censoring and Truncation

Censoring Schemes and Likelihood Construction

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1 Right Censoring

Definition 4.1: Right-censored data

X is only known to exceed some value. With event time X and censoring time C we observe

$$(T, \delta), \qquad T = \min(X, C), \qquad \delta = I(X < C) \text{ (or } I(X \leq C)),$$

for $i = 1, \ldots, n$.

Some types of right censoring give information on C even when $X < C$.

1.1 Type I censoring

The event is observed only before a *pre-specified* time.

- **Fixed Type I**: $C_i = \tau$ for all i, e.g. mice followed for a fixed period after carcinogen exposure, survivors censored at study end.
- **Progressive Type I**: C fixed in advance with finitely many distinct values, e.g. time to a given weight loss with mice in cages 1–2 sacrificed after week 1 and cages 3–4 after week 2.

- **Generalized Type I**: C_i individual but known in advance, e.g. *staggered entry* in a Phase I study: C_i = (end of study) − (entry time), assuming no dropout.

1.2 Type II censoring

- **Simple Type II**: continue until the first r of n failures; r and $n-r$ are *fixed*, $C = X_{(r)}$ is *random*.
- **Progressive Type II**: after r_1 failures remove $n_1 - r_1$ of the $n - r_1$ survivors; after the next r_2, remove $n_2 - r_2$; and so on.

Remark 4.2: Type I vs. Type II

Type I: C predetermined. Type II: C is the random time of a predetermined number of events. This matters for the likelihood.

1.3 Random censoring

Human studies: $C \sim G(t)$, not fixed, possibly unknown.

- Administrative censoring at study end (like Type I).
- Drop-out before study end (random).
- In practice, a mixture.

2 Left, Double and Interval Censoring

Definition 4.3: Left censoring

X is only known to be less than C_l; exact iff $X \geq C_l$. Data: $T = \max(X, C_l)$, $\delta = I(X \geq C_l)$.

Definition 4.4: Double censoring

With left censoring time C_l and right censoring time C_r $(C_l < C_r)$:

$$T = \max\{\min(X, C_r),\, C_l\}, \qquad \delta = \begin{cases} 1, & C_l \leq X \leq C_r \text{ (exact)} \\ -1, & X < C_l \text{ (left censored)} \\ 0, & X > C_r \text{ (right censored)} \end{cases}$$

Remark 4.5: Double vs. interval censoring

Interval: each observation lies in an interval $L < X \leq R$, none exact. **Double**: each is exact, right or left censored.

3 Truncation

Definition 4.6: Truncation

Only event times in a window (Y_L, Y_R) are observed:

- $Y_R = \infty$: **left truncation** ($Y_L < X$);
- $Y_L = 0$: **right truncation** ($X \le Y_R$).

Inference is *conditional*.

Example 4.7: Breast cancer onset (classification exercise)

8 healthy 50-year-old women had yearly mammograms for 10 years. Onset ages:

$$(55, 56],\ (58, 59],\ (52, 53],\ (59, 60],\ >60,\ >60,\ >60,\ >60.$$

Answer. Four detected cases are **interval censored**, four undetected ones **right censored** at 60, and all are **left truncated** at 50 (only cancer-free women enrolled).

4 Likelihood Construction

4.1 Contributions of each observation type

X continuous with f, F, S; C_r, C_l censoring points, $(L, R]$ censoring interval, Y truncation point.

Observation	Likelihood contribution
Exact lifetime	$f(x)$
Right censored	$S(C_r) = P(X > C_r)$
Left censored	$1 - S(C_l) = P(X < C_l)$
Interval censored	$S(L) - S(R) = P(L < X \le R)$
Left truncated	$f(x)/S(Y_L)$
Right truncated	$f(x)/[1 - S(Y_R)]$
Interval truncated	$f(x)/[S(Y_L) - S(Y_R)]$

Truncated contributions are conditional densities, e.g. $f(x \mid X > Y_L) = f(x)/S(Y_L)$.

Theorem 4.8: General censored-data likelihood

$$L \propto \prod_{i \in D} f(x_i) \prod_{i \in RC} S(C_{r,i}) \prod_{i \in LC} F(C_{l,i}) \prod_{i \in IC} \{S(L_i) - S(R_i)\},$$

with D exact times and $RC/LC/IC$ the censored observations.

Truncation. Left truncated: $f(x_i) \to f(x_i)/S(Y_i)$, $S(C_{r,i}) \to S(C_{r,i})/S(Y_i)$. Right truncated: $f(x_i) \to f(x_i)/[1 - S(Y_i)]$.

4.2 Type I right censoring

Theorem 4.9

Observing (T_i, δ_i) with $T_i = \min(X_i, C_i)$, $\delta_i = I(T_i = X_i)$ and fixed C_i,

$$L = \prod_{i=1}^{n} \Pr(t_i, \delta_i) = \prod_{i=1}^{n} f(t_i)^{\delta_i} S(t_i)^{1-\delta_i} = \prod_{i=1}^{n} h(t_i)^{\delta_i} e^{-H(t_i)}.$$

Proof [Supplement]. With C_i fixed, $\delta_i = 0$ means $X_i > C_i = t_i$, probability $S(t_i)$; $\delta_i = 1$ gives density $f(t_i)$. With $f = hS$, $S = e^{-H}$: $f^{\delta} S^{1-\delta} = (hS)^{\delta} S^{1-\delta} = h^{\delta} S = h^{\delta} e^{-H}$. ■

Example 4.10: Exponential with right censoring

$X \sim \text{Exp}(\lambda)$, $T = \min(X, C)$, $\delta = I(X < C)$:

T_i	0.5	1	0.75	0.25	0.75
δ_i	1	1	1	0	0

With $h(t) = \lambda$ and $H(t) = \lambda t$,

$$L(\lambda) = \prod_{i=1}^{5} \lambda^{\delta_i} e^{-\lambda t_i} = \lambda^{\sum \delta_i} e^{-\lambda \sum t_i} = \lambda^3 e^{-3.25\lambda}.$$

$\ell'(\lambda) = 3/\lambda - 3.25 = 0$ gives $\hat{\lambda} \approx 0.923$; generally $\hat{\lambda} = \sum \delta_i / \sum t_i$ (events / time at risk).

4.3 Type II censoring

Theorem 4.11

With n i.i.d. $X_i \sim f$, observing the r smallest order statistics $X_{(1)} < \cdots < X_{(r)}$ and $n - r$ values censored at $X_{(r)}$,

$$L = \frac{n!}{(n-r)!} \prod_{i=1}^{r} f(x_{(i)}) \left\{S(x_{(r)})\right\}^{n-r}.$$

Derivation: where the constant comes from

$n!/(n-r)!$ ordered choices of the r failing units, each contributing $f(x_{(i)})$, and $n-r$ survivors beyond $x_{(r)}$. The constant is parameter-free, so the kernel matches Theorem 4.8.

Progressive Type II (two stages: r_1 failures, $n_1 - r_1$ removed, r_2 failures $X^*_{(i)}$, the rest removed):

$$L \propto \prod_{i=1}^{r_1} f(x_{(i)}) \left\{S(x_{(r_1)})\right\}^{n_1 - r_1} \prod_{i=1}^{r_2} f(x^*_{(i)}) \left\{S(x^*_{(r_2)})\right\}^{n - n_1 - r_2} \propto \prod_{i \in D} f(x_i) \prod_{i \in RC} S(C_i).$$

4.4 Random censoring

$X_i \sim f, S$ and $C_i \sim g, G$ random and **independent**; $T_i = \min(X_i, C_i)$, $\delta_i = I(X_i \le C_i)$.

Theorem 4.12

$$L = \prod_{i=1}^{n} \{f(t_i)G(t_i)\}^{\delta_i} \{g(t_i)S(t_i)\}^{1-\delta_i} = \underbrace{\prod_{i=1}^{n} f(t_i)^{\delta_i} S(t_i)^{1-\delta_i}}_{L_X} \underbrace{\prod_{i=1}^{n} g(t_i)^{1-\delta_i} G(t_i)^{\delta_i}}_{L_C}.$$

Proof [Supplement]. $P(T \in \mathrm{d}t, \delta = 1) = P(X \in \mathrm{d}t,\ C \geq t) = f(t)G(t)\,\mathrm{d}t$ and $P(T \in \mathrm{d}t, \delta = 0) = g(t)S(t)\,\mathrm{d}t$ by independence. ■

If the law of C is free of θ (**non-informative censoring**), L_C is constant and

$$L \propto L_X = \prod_{i=1}^{n} f(t_i)^{\delta_i} S(t_i)^{1-\delta_i},$$

as under Type I censoring.

5 Remarks

- Used mainly for parametric models (K&M Ch. 12) and behind the Cox *partial likelihood* (Ch. 8–9).
- Rarely written explicitly, but Theorem 4.8 explains each data component's contribution.

6 Selected exercises

Motivation. K&M Section 3.7: a likelihood with truncation and two kinds of censoring, and the exact law of the exponential MLE under random censoring.

Exercise 4.1 (K&M 3.2: periodic exams with delayed entry)

Disease-free women enrolled from 1970 were followed 30 years for age X at breast cancer onset, examined every 3 years (onset detected only at exams). Classify, and give the likelihood contribution:
(a) enrolled at age 30, never developed breast cancer during the study;
(b) enrolled at age 40, diagnosed at the fifth exam after enrollment;
(c) enrolled at age 50, died of an unrelated cause at age 61 without a diagnosis;
(d) enrolled at age 42, moved away at age 55 without a diagnosis.
(e) Write the likelihood for these four women.

Solution. Each woman is *left truncated* at entry age v (contribution divided by $S(v)$); onset is known only up to exams at $v+3, v+6, \ldots$

(a) Exams $33, \ldots, 60$ negative: right censored at 60; $S(60)/S(30)$.

(b) Exam 4 (age 52) negative, exam 5 (55) positive: interval censored; $\{S(52) - S(55)\}/S(40)$.

(c) Exams $53, 56, 59$ negative; death at 61 ends follow-up, onset could be unseen after 59: right censored at 59; $S(59)/S(50)$. (If onset before death would surely be noticed, $S(61)/S(50)$.)

(d) Exams $45, \ldots, 54$ negative, then lost: $S(54)/S(42)$.

(e) With independent censoring and truncation,

$$L = \frac{S(60)}{S(30)} \cdot \frac{S(52) - S(55)}{S(40)} \cdot \frac{S(59)}{S(50)} \cdot \frac{S(54)}{S(42)}.$$

■

Exercise 4.2 (K&M 3.8: exponential lifetimes, exponential censoring)

Let $X \sim \text{Exp}(\lambda)$ and $C \sim \text{Exp}(\theta)$ be independent, $T = \min(X, C)$, $\delta = I(X \le C)$.
(a) Find $P(\delta = 1)$.
(b) Find the distribution of T.
(c) Show that δ and T are independent.
(d) For a random sample (T_i, δ_i), $i = 1, \ldots, n$, show that the MLE of λ is $\hat{\lambda} = \sum_i \delta_i / \sum_i T_i$, and use (a)–(c) to find its mean ($n \ge 2$) and variance ($n \ge 3$).

Solution.

(a)–(c) For $t \ge 0$,

$$P(\delta = 1,\, T > t) = P(t < X \le C) = \int_t^\infty \lambda e^{-\lambda s} e^{-\theta s}\, \mathrm{d}s = \frac{\lambda}{\lambda + \theta} e^{-(\lambda+\theta)t}.$$

$t = 0$ gives $P(\delta = 1) = \lambda/(\lambda + \theta)$; $P(T > t) = e^{-(\lambda+\theta)t}$, so $T \sim \text{Exp}(\lambda + \theta)$; and $P(\delta = 1, T > t) = P(\delta = 1)P(T > t)$ (hence also for $\delta = 0$): $\delta \perp\!\!\!\perp T$.

(d)

Step 1 (MLE). $L = \lambda^D e^{-\lambda G} \cdot \theta^{n-D} e^{-\theta G}$, $D = \sum \delta_i$, $G = \sum T_i$; $\partial_\lambda \log L = D/\lambda - G$ changes sign at D/G (if $D = 0$, $\hat{\lambda} = 0$).

Step 2 (distributions). By (c) all δ_i, T_i are independent: $D \sim \text{Bin}(n, p)$, $p = \lambda/r$, $r = \lambda + \theta$; $G \sim \text{Gamma}(r, n)$ (shape n, rate r); $D \perp\!\!\!\perp G$.

Step 3 (inverse moments of G). For $k < n$,

$$\text{E}(G^{-k}) = \int_0^\infty g^{-k} \frac{r^n g^{n-1} e^{-rg}}{\Gamma(n)}\, \mathrm{d}g = \frac{r^k\, \Gamma(n-k)}{\Gamma(n)},$$

so $\text{E}(G^{-1}) = r/(n-1)$ and $\text{E}(G^{-2}) = r^2/\{(n-1)(n-2)\}$.

Step 4 (mean). $\text{E}\hat{\lambda} = np \cdot r/(n-1) = n\lambda/(n-1)$, slightly biased upward.

Step 5 (variance). $\text{E}(D^2) = np(1-p) + n^2p^2$, $r^2p(1-p) = \lambda\theta$, $r^2p^2 = \lambda^2$:

$$\text{E}\hat{\lambda}^2 = \frac{n\lambda\theta + n^2\lambda^2}{(n-1)(n-2)},$$

$$\text{Var}\hat{\lambda} = \frac{n\lambda\theta + n^2\lambda^2}{(n-1)(n-2)} - \frac{n^2\lambda^2}{(n-1)^2} = \frac{n\lambda\theta}{(n-1)(n-2)} + \frac{n^2\lambda^2}{(n-1)^2(n-2)}.$$

As $n \to \infty$, $n\text{Var}\hat{\lambda} \to \lambda^2/p$, the inverse Fisher information per observation. ■